Claims

1. A frequency modulated continuous wave, FMCW, synthesizer (208), comprising:

a phase locked loop, PLL, (302) including:

a frequency divider (310) including a control input, a frequency divider input, and a frequency divider output, and configured to modify a frequency of a signal, received by the frequency divider input, in response to the control input;

a control voltage generator, CVG, including a first CVG input, a second CVG input, and a control voltage output, the first CVG input adapted to receive a frequency reference signal (308), the second CVG input coupled to the frequency divider output, the CVG configured to generate a control voltage (VCTRL) in response to the first CVG input and the second CVG input; and

a voltage controlled oscillator, VCO, (330) including a VCO input and a VCO output, the VCO input coupled to the control voltage output, the VCO output coupled to the frequency divider input, the VCO configured to output a FMCW output signal having a FMCW output frequency in response to the VCO input; and

a control engine (304) including a control output coupled to the control input of the frequency divider (310), the control engine (304) configured to control the PLL (302) so that the FMCW output frequency:

from a first time (t0) to a second time (t1), is a first idle frequency (F_0);

from the second time (t1) to a third time (t2), changes at a specified first slope (s1);

from the third time (t2) to a fourth time (t3), changes at a specified second slope (s2), wherein the first slope (s1) is different from the second slope (s2), wherein an absolute value of the second slope (s2) is greater than an absolute value of the first slope (s2); and

from the fourth time (t3) to a fifth time (t4), is a second idle frequency.

2. The FMCW synthesizer of claim 1, further comprising:

a receiver adapted to receive a reflected FMCW chirp, the receiver including a receiver output;

an analog to digital converter (ADC) including an ADC input and an ADC output, the ADC

input coupled to the receiver output, the ADC configured to sample signals received by the ADC input and to output the sampled signals;

a processor including a processor input coupled to the ADC output, the processor configured to, in response to the sampled signals sampled between the second time and the third time, determine at least one of presence, distance, or velocity of target objects, and configured not to perform the determine action between the third time and the fourth time; and

a transmitter including a transmitter input and configured to transmit signals received by the transmitter input, the transmitter input coupled to the VCO output.

3. The FMCW synthesizer of claim 1,

wherein the control output is a first control output of the control engine;

wherein the control engine includes a second control output and a third control output;

wherein the CVG includes:

a charge pump (CP) including a CP control input and a CP output, the CP control input coupled to the second control output, wherein a current of the CP output is configured to change in response to the CP control input; and

a filter that includes a filter signal input, a filter control input, and a filter output, the filter signal input coupled to the CP output, and the filter control input coupled to the third control output, wherein the filter output is the CVG output, and wherein a bandwidth of the filter is configured to change in response to the filter control input.

4. The FMCW synthesizer of claim 3,

wherein the control engine is configured to a) control the bandwidth of the filter to be relatively lower between the second time and the third time; and

wherein the control engine is configured to control the bandwidth of the filter to be relatively higher between the third time and the fourth time; optionally wherein the filter is configured to change the bandwidth of the filter by changing one or more impedances of the filter; or b) the current of the CP output to be relatively lower between the second time and the third time; and

wherein the control engine is configured to control the current of the CP output to be relatively higher between the third time and the fourth time.

5. The FMCW synthesizer of claim 1, wherein the first slope is a constant and the second slope is a constant.

6. The FMCW synthesizer of claim 1,

wherein the control output is a first control output of the control engine;

wherein the control engine includes a second control output;

wherein the CVG includes a CVG control input coupled to the second control output; and

wherein the CVG is configured to change a current of the control voltage output in response to the CVG control input;

wherein the control engine is configured to control the current of the control voltage output to be relatively lower between the second time and the third time; and

wherein the control engine is configured to control the current of the control voltage output to be relatively higher between the third time and the fourth time.

7. The FMCW synthesizer of claim 1,

wherein the first FMCW idle frequency is different from the second FMCW idle frequency; and

wherein the control engine is configured to control the PLL so that the FMCW output frequency:

from the fifth time to a sixth time or vice versa, changes at a specified third rate;

from the sixth time to a seventh time or vice versa, changes at a specified fourth rate, wherein the third rate is different from the fourth rate; and

from the seventh time to an eighth time or vice versa, is a third FMCW idle frequency.

8. The FMCW synthesizer of claim 1, wherein a time order either starts with the first time, then continues to the second time, then the third time, then the fourth time, then the fifth time, or starts with the fifth time, then continues to the fourth time, then the third time, then the second time, then the first time.

9. The FMCW synthesizer of claim 1, wherein the control engine includes a feedback input, the frequency divider output is coupled to the feedback input, and the control engine controls the PLL in response to the feedback input.

10. A frequency modulated continuous wave (FMCW) radar system, comprising:

a receiver adapted to receive a reflected FMCW chirp, the receiver including a receiver output;

a FMCW synthesizer (208) according to any of claims 1 to 9, including:

a FMCW synthesizer output;

the phase locked loop (PLL) including the control input and a PLL output coupled to the FMCW synthesizer output, and configured to output a FMCW signal with the FMCW output frequency in response to the control input; and

the control engine including a feedback input and the control output, the feedback input coupled to the FMCW synthesizer output, the control output coupled to the control input, the control engine configured to control the PLL in response to the feedback input so that the FMCW output frequency:

from the first time to the second time, is a first FMCW idle frequency;

from the second time to the third time, changes at a specified first slope (s1);

from the third time to the fourth time, changes at a specified second slope (s2), wherein the first slope (s1) is different from the second slope (s2), wherein an absolute value of the second slope (s2) is greater than an absolute value of the first slope (s2); and

from the fourth time to the fifth time, is a second FMCW idle frequency;

a mixer including a first mixer input, a second mixer input, and a mixer output, the first mixer input coupled to the receiver output, the second mixer input coupled to the frequency synthesizer output, the mixer configured to generate mixer output signals in response to the first mixer input and the second mixer input;

an analog to digital converter (ADC) including an ADC input and an ADC output, the ADC input coupled to the mixer output, the ADC configured to sample signals received by the ADC

input and to output the sampled signals;

a processor including a processor input coupled to the ADC output, the processor configured to, in response to the sampled signals sampled between the second time and the third time, determine at least one of presence, distance, or velocity of target objects, and configured not to perform the determine action between the third time and the fourth time; and

a transmitter including a transmitter input and configured to transmit signals received by the transmitter input, the transmitter input coupled to the FMCW synthesizer output.

11. The FMCW synthesizer of claim 1 or the FMCW radar system of claim 10, wherein the control engine is configured to control the PLL to maintain phase lock both when generating FCMW chirps and during idle times between generation of FMCW chirps.

12. The FMCW radar system of claim 10, wherein the control engine is configured to control the PLL to:

increase a filter bandwidth of a filter of the PLL during idle times between generation of FMCW chirps;

lower the filter bandwidth of the filter of the PLL during generation of FMCW chirps;

increase a current of a control voltage of a voltage controlled oscillator (VCO) of the PLL during the idle times between generation of FMCW chirps; and

lower the current of the control voltage of the VCO of the PLL during generation of the FMCW chirps.

13. A method for operating a frequency modulated continuous wave (FMCW) radar, comprising:

receiving, by a receiver, a FMCW chirp reflection signal;

generating, using a FMCW synthesizer, a FMCW synthesizer signal at a FMCW synthesizer output frequency, wherein the FMCW synthesizer output frequency:

from a first time to a second time, is a first FMCW idle frequency;

from the second time to a third time, changes at a specified first slope;

from the third time to a fourth time, changes at a specified second slope, wherein

the first slope is different from the second slope, wherein an absolute value of the second slope (s2) is greater than an absolute value of the first slope (s2); and

from the fourth time to a fifth time, is a second FMCW idle frequency;

mixing, using a mixer, the FMCW chirp reflection signal and the FMCW synthesizer signal, to generate a mixer output;

sampling the mixer output using an analog-to-digital converter (ADC); and

transmitting, using a transmitter, the FMCW synthesizer signal.

14. The method of claim 13, further comprising:

lowering a bandwidth of a filter of a phase locked loop (PLL) of the FMCW synthesizer to be relatively lower between the second time and the third time; and

raising the bandwidth of the filter of the PLL of the FMCW synthesizer to be relatively higher between the third time and the fourth time.

15. The method of claim 13, further comprising:

lowering a current of an output of a charge pump of a phase locked loop (PLL) of the FMCW synthesizer to be relatively lower between the second time and the third time; and

raising a current of the output of the charge pump of the PLL of the FMCW synthesizer to be relatively higher between the third time and the fourth time.

16. The method of claim 13,

wherein the first FMCW idle frequency is different from the second FMCW idle frequency; and

further comprising generating, using the FMCW synthesizer, the FMCW synthesizer signal, wherein the FMCW synthesizer output frequency:

between the fifth time and a sixth time, changes at a specified third rate;

between the sixth time and a seventh time, changes at a specified fourth rate, wherein the third rate is different from the fourth rate; and

between the seventh time and an eighth time, is a third FMCW idle frequency.

17. The method of claim 13, further comprising maintaining phase lock, using a phase locked loop (PLL) of the FMCW synthesizer, both when generating FCMW chirps and during idle times between generation of FMCW chirps.

18. The method of claim 13, wherein a time order either starts with the first time, then continues to the second time, then the third time, then the fourth time, then the fifth time, or starts with the fifth time, then continues to the fourth time, then the third time, then the second time, then the first time.